

1) A sample of men agreed to participate in a study to determine the relationship between several variables including height, weight, waste size, and percent body fat. A scatterplot with percent body fat on the y-axis and waist size (in inches) on the horizontal axis revealed a positive linear association between these variables. Computer output for the regression analysis is given below:

Dependent variable is: %BF

R-squared = 67.8%

S = 4.713 with 250-2 = 248 degrees of freedom

Variable	Coefficient	se of coeff	t-ratio	prob
Constant	-42.734	2.717	-15.7	<.0001
Waist	1.70	0.0743	22.9	<.0001

(a) Write the equation of the regression line:

(b) Explain/interpret the information provided by R-squared in the context of this problem. Be specific.

(c) One of the men who participated in the study had waist size 35 inches and 10% body fat. Calculate the residual associated with the point for this individual.

2) Scientists are trying to study the relationship between the weight of an animal's heart and the length of the cavity of the heart's left ventricle. The following data was collected from various animals

Length of cavity of left ventricle (in cm)	Heart weight (in grams)
.55	.13
1.0	.64
2.2	5.8
4.0	102
6.5	210
12.0	2030
16.0	3900

a) Make a scatter plot in the space above and confirm that the relationship does not appear to be linear.

b) Based on the ladder of transformations come up with a few possible transformations and test them out by:

- i. Visually inspecting the transformed scatter plot
- ii. Looking at the residual plots of the LSRL of the transformed data
- iii. Comparing the correlation coefficients of the various choices

c) Write out the transformed regression equation.

d) An animal was found to have a left ventricle cavity of length 10cm. What would be your projected weight for the animal's heart?

e) Would it make sense to try to predict the weight of a heart using our equation if the length of the ventricle is 20cm? Why?

_____ 3) It's easy to measure the circumference of a tree's trunk but not so easy to measure its height. Foresters developed a model for ponderosa pines that they use to predict the tree's height (in feet) from the circumference of its trunk (in inches): $\ln \hat{y} = -1.2 + 1.4 \ln c$. A lumberjack finds a tree with a circumference of 60"; how tall does this model estimate the tree to be?

- (A) 19' (B) 5' (C) 11' (D) 93' (E) 83'